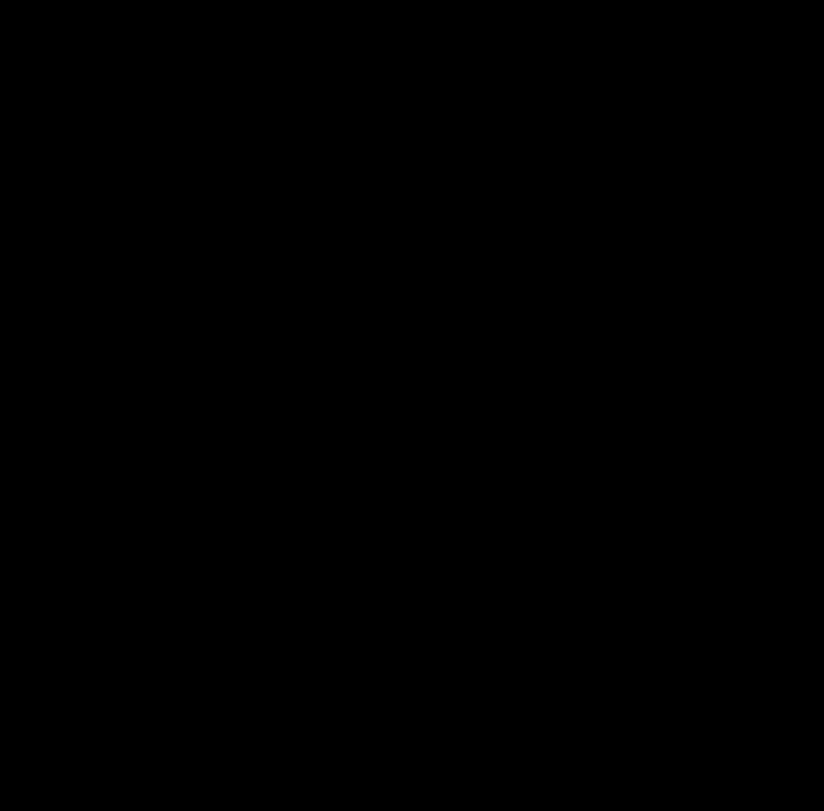




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INSTRUCTION TO CANDIDATE:

1. This Exam paper is composed of **three sections (A, B and C)**. Follow the instructions given below and answer the indicated questions for a total of **100 marks**.

Section **A**: All questions are **Compulsory** **55 marks**

Section **B**: Attempt any three (3) **30 marks**

Section **C**: Attempt anyone (1) **15 marks**

2. Fill in your actual **names** and **Index number** on the provided space(Cover).

3. Do not remove or tear any page or part of this Questions and Answers Booklet.

4. Answer in the language in which the Examination is set.

5. In section **A** answer questions in the provided space. In case of multiple choice questions, just **circle** the letter corresponding to the correct option.

6. In sections **B** and **C**, answer questions in the provided space after section C questions.

7. **You do not need the Periodic Table.**

8. **Allowed materials:** - Blue or Black pen.

- Geometrical Instruments

- Silent non-programmable calculators

T2-174_Applied Chemistry C

SECTION A: Attempt all questions

(55 marks)

- 01.** Fill in the blank spaces, of the following sentences, with the provided underlined terms. Each term will be used once: (***sub-atomic, neutrons, neutral***).
- i) An atom is electrically because of equal number of positively charged protons and negatively charged electrons. **(1Mark)**
 - ii) Protons and.....are found inside the nucleus. **(1Mark)**
 - iii) There are three main..... particles in the atom. **(1Mark)**
- 02.** Circle a letter corresponding to the correct answer to each of the following sub-questions.
- i) What does a burette used for in laboratory setting? **(1Mark)**
 - A) To measure and transfer large volumes accurately.
 - B) To heat substances to high temperature.
 - C) To measure and dispense precise volume of liquid during titration.
 - D) To mix chemical solutions thoroughly.
 - ii) What does acidimetry mean among the following descriptions? **(1Mark)**
 - A) A technique for determining the concentration of an acid in a solution.
 - B) A method used for to measure the basicity of a solution.
 - C) The process of neutralizing an acid a base with a base.
 - D) A method for measuring the pH level of a solution.
 - iii) Among the following options which one correctly define the term alkalimetry? **(1Mark)**
 - A) A method used for to measure the acidity of a solution.
 - B) A technique used for determining the concentration of a base in a solution.
 - C) The process of neutralizing an acid a base with a base.
 - D) A method for measuring the pH level of a solution.

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03. Using True(**T**) or False (**F**) state whether each of the following statements about organic compounds are correct or wrong.

- i) The general molecular formula for alkanes is C_nH_{2n+2} (**1Mark**)
- ii) Organic compounds generally have higher melting points than inorganic compounds. (**1Mark**)
- iii) Organic compounds are typically less volatile than inorganic compounds. (**1Mark**)

04. Match using middle column each class of carbohydrates in the left column, with its corresponding example in the right column of the table. (**3Marks**)

Class of carbohydrates	Answer	Example
1) Monosaccharides	1= ...	A) Sucrose
2) Disaccharides	2=	B) Starch
3) Polysaccharides	3=	C) Glucose

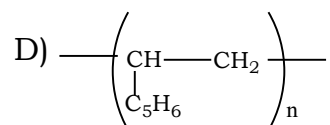
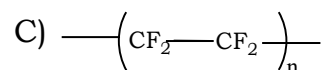
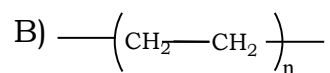
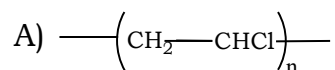
05. Circle the letter corresponding to the correct answer to each of the following questions.

- i) What is a monomer? (**1Mark**)
- A) A large molecule made up of many repeating subunits.
- B) A single, small molecule that can combine with others to form a polymer.
- C) A type of lipid used in cell membrane.
- D) A complex protein that catalyzes reactions in living organisms.
- ii) Which of the following polymers is a synthetic condensation polymer? (**1Mark**)
- A) Polyethene
- B) Polystyrene
- C) Polyester
- D) Polypropene

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iii) Which of the following repeating units is for polyvinylchloride (PVC)?

(1Mark)



06. Circle the letter corresponding to the correct answer to each of the following questions.

i) Which of the following chemicals is an example of a silicate?

(1Mark)

- A) NaCl
- B) CaCO₃
- C) MgSO₄
- D) K₂SiO₃

ii) Which of the following alloys is called bronze?

(1Mark)

- A) Alloy of iron + carbon
- B) Alloy of iron + carbon + chromium
- C) Alloy of copper + tin
- D) Alloy of copper + zinc

iii) Which of the following processes is an example of freezing as a phase transition?

(1Mark)

- A) Liquid water turning into ice
- B) Liquid water turning into steam
- C) Ice turning into water
- D) Oxygen turning into liquid

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07. Fill the blank spaces of the following sentences, with appropriate terms from the list provided: (**nitrogen fixation, carbon cycle, denitrification, sulphur cycle**).

- i) The process by which carbon is exchanged between the atmosphere, oceans, soil, and living organisms is known as the **(1Mark)**
- ii) In the nitrogen cycle,
is the process in which atmospheric nitrogen is converted into ammonia by bacteria in the soil or root nodules of plants. **(1Mark)**
- iii) The involves the movement of sulfur through the atmosphere, lithosphere, and hydrosphere, including processes like sulfur emissions from volcanoes. **(1Mark)**
- iv) The process of occurs when organisms, such as bacteria, convert nitrate back into nitrogen gas (N_2) in the nitrogen cycle. **(1Mark)**

08. Circle the letter corresponding to the correct answer to each of the following sub-questions.

- i) Which two (2) elements of the following can help to find the position of an element in the periodic table? **(2Marks)**
- A) Atomic number
B) Atomic mass
C) Electron configuration
D) Number of neutrons
- ii) Why is an atom said to be electrically neutral? **(2Marks)**
- A) An atom has more protons than electrons.
B) The number of protons is equal to the number of electrons and their charges cancel each other.
C) The number of neutrons is equal to the number of electrons and their charges cancel each other.

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09. Use the column in the middle with number given to chemical equation and match it to the roman number given to the type of reaction in the table below. (4Marks)

Chemical reaction	Answer	Type of reaction
1. $\text{C(s)} + \text{O}_2\text{(g)} \longrightarrow \text{CO}_2\text{(g)}$	1.=	i. Decomposition reaction
2. $\text{HCl(aq)} + \text{KOH(aq)} \longrightarrow \text{KCl(aq)} + \text{H}_2\text{O(aq)}$	2.=	ii. Combustion reaction
3. $\text{CaCO}_3\text{(aq)} \longrightarrow \text{CaO(s)} + \text{CO}_2\text{(g)}$	3.=	iii. Precipitation reaction
4. $\text{AgNO}_3\text{(aq)} + \text{NaCl(aq)} \longrightarrow \text{AgCl(s)} + \text{NaNO}_3\text{(aq)}$	4.=	iv. Neutralization reaction

10. Using **True(T)** or **False (F)** state whether each of the following statements about organic compounds are correct or wrong.

- i) Silicon carbide (SiC) is an example of organic compound (1Mark)
- ii) Alicyclic organic compounds are compounds with open carbon chains only. (1Mark)
- iii) Generally, organic compounds are insoluble in water. (1Mark)
- iv) The compound, $\text{CH}_3\text{CH}_2\text{CHNH}_2$ is an example of hydrocarbon compound. (1Mark)

11. Fill the blank spaces in the following sentences, with appropriate terms provided as: (**insulin, nucleotides, amino acids, ribonucleic acid**).

- i) The nucleic acids are macromolecules composed of (1Mark)
- ii) An example of nucleic acid is (1Mark)
- iii) Proteins are large macromolecules that are made by condensation of (1Mark)
- iv) An example of protein is..... (1Mark)

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12. Match each process in the left column, with its corresponding meaning in the right column of the table below.

(4Marks)

Process	Answer	Meaning
1.Absorption	(1) =	A). Type of water that contains high levels of dissolved minerals, mainly calcium (Ca^{2+}) and magnesium (Mg^{2+}).
2.Adsorption	(2) =.....	B). Process in which one substance is taken up into the interior of another substance.
3.Hard water	(3) =	C). Process in which a substance adheres to the surface of another material.
4.Soft water	(4) = ...	D). Type of water that contains low levels of calcium (Ca^{2+}) and magnesium (Mg^{2+}), often due to the presence of sodium and potassium ions.

13. Circle the letter corresponding to the correct answer to each of the following sub-questions. An atom of element **Y** has the following representation ${}^{37}_{17}\text{Y}$.

i) How many protons does **Y** have?

(1Mark)

- A) 17 protons
- B) 20 protons
- C) 37 protons
- D) 54 protons

ii) How many electrons does **Y** have?

(1Mark)

- A) 54 electrons
- B) 37 electrons
- C) 20 electrons
- D) 17 electrons

iii) How many neutrons does **Y** have?

(1Mark)

- A) 54 neutrons
- B) 37 neutrons
- C) 20 neutrons
- D) 17 neutrons

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iv) What is the correct electron configuration for ${}^{37}_{17}\text{Y}$?

(1Mark)

- A) $1s^2 2s^2 3p^6 3s^2 3p^6 4s^2$
- B) $1s^2 2s^2 3p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^1$
- C) $1s^2 2s^2 3p^6 3s^2 3p^5$
- D) $1s^2 2s^2 3p^6 3s^2 3p^6$

14. Circle the letter corresponding to the correct answer to each of the following sub-questions.

i) Which two (2) of the following substances give neutral solutions, with pH=7?

(2Marks)

- A) Nitric acid (HNO_3)
- B) Sodium chloride (NaCl)
- C) Potassium hydroxide (KOH)
- D) Sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$)

ii) What is the molarity of a solution made by adding 11 g of sodium chloride (NaCl) to make 250 ml of solution?

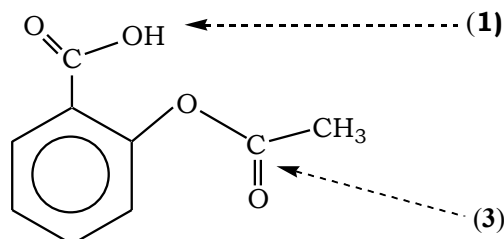
(2Marks)

- A) 0.188 mol/l
- B) 0.044 mol/l
- C) 44 mol/l
- D) 0.752 mol/l

15. The structure of aspirin, a pharmaceutical compound used as pain relieve and anti-inflammatory, structure is shown below.

(circle the letter(s) corresponding to the right answer)

Aspirin



i) To which class of organic compounds does aspirin belong?

(2Marks)

- A) Hydrocarbon compound
- B) Aliphatic compound

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C) Aromatic compound

D) Alicyclic compound

ii) Name the three functional groups found in aspirin.

a) (1) = (2Marks)

b) (2) = (2Marks)

c) (3) = (2Marks)

Section B: Attempt any three (3) questions

(30 marks)

16. Circle the letter corresponding to the correct answer to each of the following sub-questions.

i) What is meant by oxidation state of an element in a compound? (1Mark)

A) The total number of electrons in an atom's outermost shell.

B) The number of electrons an atom has lost or gained in a chemical combination.

C) The number of bonds an atom has formed in a compound.

D) The total number of electrons in an atom's inner shell.

ii) What is the oxidation state of manganese (Mn) in pyrolusite (MnO_2)? (2Marks)

A) +2

B) +3

C) +4

D) +6

iii) In the reaction, $\text{Cu}_2\text{O}(\text{s}) + \text{H}_2\text{SO}_4(\text{aq}) \longrightarrow \text{Cu}(\text{s}) + \text{CuSO}_4(\text{aq}) + \text{H}_2\text{O}$

, which element has undergone both oxidation and reduction? (2Marks)

A) Copper (Cu)

B) Hydrogen (H_2)

C) Sulphur (S)

D) Oxygen (O_2)

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iv) In the reaction, $\text{H}_2\text{S} + \text{FeCl}_3 \longrightarrow \text{HCl} + \text{S} + \text{FeCl}_2$,

which compound acts as a reducing agent?

(2Marks)

- A) Hydrogen sulphide (H_2S)
- B) Iron trichloride (FeCl_3)
- C) Hydrogen chloride (HCl)
- D) Iron dichloride (FeCl_2)

v) What is the correct set of stoichiometric coefficients in the redox equation $x \text{H}_2\text{S} + y \text{FeCl}_3 \longrightarrow z \text{HCl} + w \text{S} + v \text{FeCl}_2$?

(3Marks)

- A) $x = 1, y = 1, z = 2, w = 1$ and $v = 1$
- B) $x = 3, y = 2, z = 1, w = 1$ and $v = 1$
- C) $x = 2, y = 1, z = 2, w = 3$ and $v = 1$
- D) $x = 1, y = 2, z = 2, w = 1$ and $v = 2$

17. Standard solutions are prepared either by dissolving a solid chemical or by diluting the liquid stock solution.

$$[\text{Molarity} = \frac{d \times \% \times 10}{Mm}] \text{ and } [M_1V_1 = M_2V_2]$$

Where d = density, $\%$ = percentage of purity of substance, Mm = molar mass M_1, M_2, V_1 and V_2 are the molarity and volume of solution before and after dilution. *Circle the letter corresponding to the correct answer for each of the following sub-questions*

i) Which of the following solutions whose concentration is determined in a titration?

(1Mark)

- A) Titrant
- B) Indicator
- C) Analyte
- D) Solvent

ii) Which one of the following, is a best apparatus specifically used in titration?

(1Mark)

- A) Centrifuge
- B) Burette
- C) Bunsen Burner
- D) Thermometer

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- iii) A bottle of stock solution of sulphuric acid (H_2SO_4) is labeled with $d = 1.84$, $\% = 98$, and $M_m = 98\text{g/mol}$. Determine the concentration in mol/l , of this stock solution. **(3Marks)**
- A) 16.4 mol/l
B) 18.1 mol/l
C) 16.4 mol/l
D) 18.4 mol/l
- iv) What volume of a $12 \text{ mol/dm}^3 \text{ HCl}$ required to prepare 500 cm^3 of a 0.5 mol/l HCl solution? **(3Marks)**
- A) 20.83 cm^3 of HCl
B) 10.83 cm^3 of HCl
C) 41.66 cm^3 of HCl
D) 83.20 cm^3 of HCl
- v) Which two (2) of the following are malpractices that can be the source of errors when preparing solutions in laboratory? **(2Marks)**
- A) Make proper calculations methods.
B) Use glassware which is not properly cleaned.
C) Incorrect reading of the mass weighed on the balance.
D) Understand well the solution requirements.
- (Consider atomic masses: $\text{H}=1$, $\text{O}=16$, $\text{S}=32$, $\text{Cl}=35.5$)

- 18.** Consider the extraction of iron from iron ore (Fe_2O_3) in the blast furnace using carbon (coke). The reaction is:



The standard enthalpies of formation of the substances are given:

$$\Delta H^\circ_f[\text{Fe}_2\text{O}_3(\text{s})] = -842 \text{ kJ/mol}$$

$$\Delta H^\circ_f[\text{Fe}(\text{l})] = -26 \text{ kJ/mol}$$

$$\Delta H^\circ_f[\text{CO}_2(\text{g})] = -393 \text{ kJ/mol}$$

Consider:

$$\Delta H_{\text{rx}} = \sum \Delta H^\circ_f(\text{products}) - \sum \Delta H^\circ_f(\text{reactants})$$

- i) Use the data given above to calculate the enthalpy for the above reaction (ΔH_{rx}). **(3Marks)**
- ii) What is the role of carbon (coke) in the blast furnace? **(2Marks)**

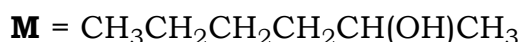
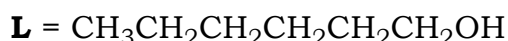
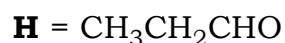
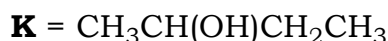
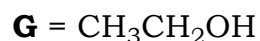
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iii) Is the reaction exothermic or endothermic? (1Mark)

iv) Mention one environmental problem that could arise during this extraction. (2Marks)

v) Suggest one method to eliminate the environmental problem stated in (iv) above. (2Marks)

19. Given a list of six organic compounds below.



i) Among the following options which one that correctly defines “Homologous series”? (2Marks)

A) A series of compounds that have no common functional group but have similar physical properties.

B) A series of compounds with similar chemical properties and some functional group differing from the successive member by $-\text{CH}_2-$ group.

C) A series of compounds that share the same functional group and differ by a single carbon atom in their structure.

D) A series of compounds with identical molecular formulae but different structural arrangements.

ii) Give the IUPAC names for the compounds **G**, **H**, **J**, and **K**. (4Marks)

iii) Identify two pairs of compounds, from the list provided before, that belong to the same homologous series. (4Marks)

20. The table below shows the standard reduction potentials for two substances.

Electrode reaction	$E^\ominus / \text{V}$
$\text{Cr}^{3+}(\text{aq}) + 3\text{e}^- \rightleftharpoons \text{Cr}(\text{s})$	- 0.74
$\text{Fe}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Fe}(\text{s})$	- 0.44

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A complete electrochemical cell $\text{Cr}^{3+}(\text{aq}) + 3\text{e}^{-} \rightleftharpoons \text{Cr}(\text{s})$ is obtained when the half-cell of chromium is connected to the half-cell of iron with all required accessories.

- i) Which of the two electrodes will be the:
- a) anode? (1Mark)
- b) cathode? (1Mark)
- ii) Write the equation that takes place at the:
- a) anode. (2Marks)
- b) cathode. (2Marks)
- iii) Calculate the cell potential for the above electrochemical cell. (2Marks)
- iv) Which of the two substances is the oxidizing agent? Justify your answer. (1Mark)

Section C: Attempt only one (1) question

(15 marks)

21. Read the following sub-questions about inorganic substances and provide appropriate answer.

- i) Match each oxide on the left, with its corresponding acid-base character on the right, in the table below. (4Marks)

Oxide	Answer	Acid-base character
A)Na ₂ O	A)=.....	(1)Acidic oxide
B)SO ₂	B)=.....	(2)Basic oxide
C)Al ₂ O ₃	C)=.....	(3)Amphoteric oxide
D)CO	D)=.....	(4)Neutral oxide

- ii) Using **True(T)** or **False (F)** state whether each of the following statements about chemical substances are correct or wrong.
- a) The pH of a 0.1 M HCl solution is equal to 13. (2Marks)
- b) The molarity of a 100 ml solution containing 1.17 g of NaCl is equal to 0.2 mol/l. (2Marks)
- c) The concentration in hydrogen ions, $[\text{H}^{+}]$, for a HNO_3 solution with pH = 2, is equal to 0.001 mol/l. (2Marks)

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d) The solution of KCl with pH = 3 has no effect neither on the red litmus paper nor the blue litmus paper.

(2Marks)

(Atomic masses: H =1, N =14, O=16, Na = 23, Cl =35.5)

iii) Use the substances, Mg, O₂, and H₂O as the only starting materials to show how you can prepare magnesium hydroxide, Mg(OH)₂(aq) through one or more steps.

(3Marks)

22. This question concerns the basics of environmental chemistry.

i) Match the methods to minimize mining pollution in the left column, with their corresponding practical examples, in the right column of the table below.

(4Marks)

Method	Answer	Practical example
A) Lower-impact Mining Technique	(A) =	1. Recovering metals from slag
B) Reusing Mining Technique	(B) =	2. In-situ Mining (Solution Mining)
C) Eco-Friendly Equipment	(C) =	3. Revegetation and Landscaping
D) Rehabilitating Mining Sites	(D) =	4. Green construction tools

ii) Which two (2) of the following are the effects of mining pollution?

(2Marks)

- A) Soil and water remediation.
- B) Greenhouse Gas Emissions (GHGs).
- C) Acid Mine Drainage (AMD).
- D) Water management.

iii) What does Environment Impact Assessment (EIA) in mining technology refer to?

(1Mark)

- A) An evaluation of the profitability of mining operations.
- B) A process to assess the financial viability of a mining project.
- C) A study that predicts the potential environmental effects of a mining project and suggests mitigation measures.
- D) A legal procedure to grant permits for mining activities.

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iv) Which of the following are two (2) correct principles of

Environment Impact Assessment (EIA) process in mining technology? **(2Marks)**

- A) Sustainability
- B) Slope stabilization
- C) Erosion control of plants
- D) Prevention and Precaution

v) Fill the blank spaces in the following sentences, with appropriate terms

from the provided list: **(climate change, chlorofluorocarbons, greenhouse effect, greenhouse gases, global warming, ozone depletion)**

- a) A group of gases that, in Earth's atmosphere trap heat by absorbing and emitting infrared radiations are called **(1Mark)**
- b) The long-term increase in Earth's average surface temperature due to human activities is known as **(1Mark)**
- c) The reduction of Earth's ozone layer, primarily by human-made chemicals **(1Mark)**
- d) The long-lasting changes in the Earth's climate, including shifts of temperature, precipitation patterns, and extreme weather is called **(1Mark)**
- e) The process by which greenhouse gases trap heat in the Earth's atmosphere, warming the planet is **(1Mark)**
- f) A group of human-made compounds composed of carbon, fluorine, and chlorine are **(1Mark)**

END

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RESERVED SPACE FOR SECTION B AND C ANSWERS

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